Unprovoked Seizures

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Introduction

- ☑ **First seizure** is a dramatic and frightening event. For parents whose child has had a first-time seizure, and for the physicians who treat them, the most salient concern after stabilizing the child is the risk of recurrence.
- ☑ **Unprovoked seizure:** Seizure or seizure clusters occurring within 24 hours in a child >1 month age occurring in the absence of precipitating factors, such as fever, hypoglycemia, dyselectrolytemia, meningitis, drug/toxin overdose, trauma, stroke, and intracranial hemorrhage.
- ☑ Incidence rates of single unprovoked seizure: 23–64.1/lakh/person-years.
- ☑ 10–12% of children and adults with a first unprovoked seizure will present with a seizure lasting ≥30 minutes (status epilepticus) as their first seizure.
- ☑ First seizure in a child can be febrile seizure (seizure associated with febrile illness but not related to neuroinfection), nonfebrile illness seizure (seizure associated with an infection but without fever like upper respiratory tract infection or gastrointestinal illness) and unprovoked seizure.
- ☑ **Acute symptomatic seizures** *(reactive or provoked seizures or situation related seizures):* These are associated with acute brain insult, which may be due to infectious, toxic, metabolic, or traumatic cause.
- ☑ **Remote symptomatic seizures:** These occur without immediate cause but with a prior identifiable major brain insult such as severe trauma or accompanying a condition such as cerebral palsy or mental retardation.
- ☑ **Genetic/idiopathic etiology:** Not associated with a known central nervous system (CNS) disorder and are of suspected genetic etiology.
- ☑ **Unknown/cryptogenic etiology:** Causes not yet known but these children also have developmental delay or other neurological deficits.

Investigations

- ☑ Serum calcium should be done in all children <2 years especially those <1 year with unprovoked seizure.
- ☑ Serum electrolytes, random blood sugar (RBS) can be obtained if the child is drowsy post seizure at presentation.
- ☑ **Electrocardiogram (ECG):** It is considered in children with seizures associated with syncope or breath-holding spell and when the event was not observed by an adult or when the event occurred while playing.
- ☑ **Electroencephalogram (EEG):** It can give clue to underlying structural lesion and guide further tests. Inclusion of both an awake and a sleep tracing, as well as hyperventilation and photic stimulation is recommended to increase the yield.
  - *Emergent EEG (<24 hours)*: It is used to confirm the occurrence of seizure and differentiate from a seizure mimic. In case of true seizures, EEG may show slowing of background (focal/generalized) and/or interictal epileptiform discharges. It also helps to diagnose nonconvulsive or subtle convulsive status epilepticus that may be mistaken for a prolonged postictal state and difficult to diagnose clinically. One-fourth of patients with treated status epilepticus who appear to be seizure-free continue to have electrographic seizure activity detectable with EEG.
  - Nonemergent EEG after 1–2 weeks can help in identifying the seizure type and epileptic syndrome.
- ☑ An EEG helps to predict recurrence, particularly if there is epileptiform activity and to help make a diagnosis of epilepsy or epileptic syndrome.
- ☑ **Neuroimaging:** Abnormal neuroimaging has been considered as a risk factor for recurrence after a first seizure.
- ☑ **Indications:**
  - Acute head trauma
  - Fever
  - History of anticoagulation/bleeding diathesis
  - History of malignancy
  - New onset or postictal focal neurologic deficit not resolving in a few hours
  - Focal seizure
  - Persistent altered mental status
  - Persistent headache
- ☑ Neuroimaging is important to exclude an epileptogenic lesion that may confer an elevated recurrence risk and in providing accurate diagnosis.
  - *Emergent*: CT has radiation risk and is inferior to MRI in identifying lesions. Migration defects and mesial temporal sclerosis, both of which are known to cause childhood onset seizures are not readily seen on CT. Hence, MRI is the preferred imaging modality.

Investigations

- *Follow-up*: Nonurgent MRI is indicated in:
  - Infants
  - Significant cognitive or motor impairment of unknown etiology
  - Unexplained abnormalities on neurologic examination
  - Focal onset of seizure
  - No evidence of benign partial epilepsy of childhood or primary generalized epilepsy on EEG.

☑ The International League Against Epilepsy (ILAE) recommended MRI when feasible for all epileptic patients, who do not have a clearly identifiable epilepsy syndrome. However, the decision is less clear after a first unprovoked seizure which may turn out to be an isolated event without any recurrence.

☑ Lumbar puncture (LP) is not routinely recommended in afebrile children. It can be considered in:

- Children <12 months
- Persistently altered mental status
- Meningeal signs
- Immunocompromised status.

If increased intracranial pressure is suspected, the LP should be preceded by neuroimaging.

Treatment

☑ **Rescue medications:** Intranasal midazolam spray 0.2 mg/kg/dose (one puff for every 5 kg—maximum 10 mg/dose), buccal midazolam (0.5 mg/kg), rectal diazepam 0.2–0.5 mg/kg (maximum 20 mg) to abort the ongoing seizure.

☑ If seizure persists for >5 minutes or if the child is seizing at presentation to emergency room, status epilepticus management guidelines can be followed: maintain airway, breathing, and circulation. Measure RBS, electrolytes, and calcium. Benzodiazepines: Injection lorazepam 0.1 mg/kg/dose or injection midazolam 0.1 mg/kg IV/IM or injection diazepam 0.2–0.3 mg/kg intravenous (IV). If seizure persists, loading with levetiracetam 40–60 mg/kg or valproate 30–40 mg/kg or phenytoin/fosphenytoin at 20 PE/kg can be given.

☑ Risks of chronic daily treatment with antiseizure medication (ASM) could be more detrimental to the child's health or development than seizure recurrence. Therefore, the risks and benefits of treatment must be individualized.

Treatment

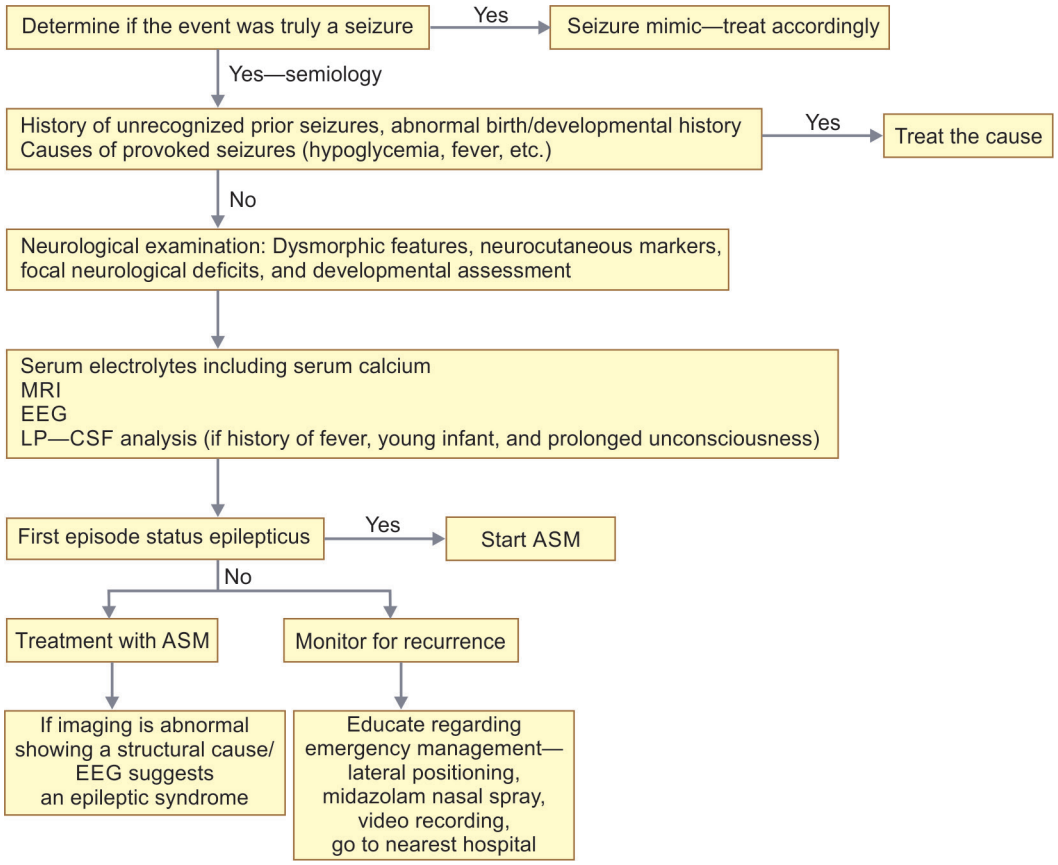
- ☑ The risk of recurrence depends on several clinical, etiological, EEG, and neuroimaging findings that should be approached on an individual basis.
- ☑ Treatment with ASM after a first seizure reduces the risk of recurrence, but does not prevent the development of epilepsy. There is no evidence of difference between starting treatment after the first seizure versus after a second seizure in achieving a 1- or 2-year seizure remission.
- ☑ Adverse effects of ASM include rash, hirsutism, weight gain, effects on behavior, and higher cortical function, which are often dose related and may be under-recognized. Severe reactions such as hepatic toxicity, bone marrow toxicity, and Stevens–Johnson syndrome cannot be anticipated and require early recognition of symptoms.
  Side effects occurred at a rate of 9% for phenytoin, 4% each for carbamazepine and valproate.
  - Initiating ASM is not generally advised, but shall be considered in individual situations where the benefits of reducing the risk of a second seizure outweigh the risks of pharmacologic and psychosocial side effects.
  - The ILAE Task Force on Sports for People With Epilepsy (PWE) based on potential risk of injury or death should a seizure occur:
    - *Group I*: Sports with no significant additional risk (athletics, bowling, most collective sports in the ground, etc.)
    - *Group II*: Sports with moderate risk to PWEs, but no risk to bystanders (gymnastics, fencing, cycling, karate, etc.)
    - *Group III*: Sports with major risk for PWE and bystanders (aviation, climbing, diving, horse riding, etc.).

Prognosis

- ☑ Most children with a first unprovoked seizure will have few or no recurrences, and about 10% will have ≥10 seizures despite therapy.
- ☑ Seizures that can occur in clusters in 24 hours associated with acute diarrheal illness with/without fever have less chance of recurrence and hence do not usually warrant long-term ASMs.
- ☑ Serious injury from a seizure in a child is a rare event, usually from a fall with loss of consciousness. To reduce that risk, restrictions are recommended that would apply to any young child, such as bicycling with a helmet on a sidewalk rather than the street and swimming only with a guardian.
- ☑ 46–54% children experience a recurrence.
- ☑ The recurrence rate by 2 years is higher in individuals who have a remote symptomatic etiology (50–80%). In those with an idiopathic/cryptogenic etiology, it is significantly lower (30–50%).

- ☑ None of the children studied prospectively had residual motor or cognitive disability.
- ☑ If a child with an initial prolonged seizure does experience a seizure recurrence, it is more likely to be prolonged.
- ☑ 10% will develop difficult-to-control and protracted epilepsy.
- ☑ Identifying genetic, immune, or imaging markers may improve prognostication.
- ☑ *Seizure recurrence rate*: 16, 21, 27, and 34% at 12, 24, 36, and 60 months, respectively.
- ☑ The majority of patients experiencing new episodes do so in the first 3 months after the initial event.
- ☑ Focal seizures have a higher risk of recurrence than generalized seizures because they are mostly associated with a structural etiology, abnormal EEG, and neuroimaging findings.

Flowchart 1: Algorithm to approach of seizures in children.



(ASM: antiseizure medication; CSF: cerebrospinal fluid; EEG: electroencephalogram; LP: lumbar puncture)

Further Reading

- ☑ Capovilla G, Kaufman KR, Perucca E, Moshe SL, Arida RM. Epilepsy, seizures, physical exercise, and sports: a report from the ILAE Task Force on Sports and Epilepsy. Epilepsia. 2016;57(1):6-12.
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- ☑ Jim'enez-Villegas MJ, Lozano Garcia L, Carrizosa-Moog. Update on first unprovoked seizure in children and adults: a narrative review. Seizure. 2021;90:28-33.

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